

Learning Objectives

1. Understand the underlying pathophysiology of shock
2. Identify the unique differences in the presentation and pathophysiology of the four major categories of shock: hypovolemic, cardiogenic, obstructive, and distributive
3. Describe the clinical approach to a patient with shock and identify the next best steps in management
4. Understand and identify the pharmacologic treatments for each of the major categories of shock

Mechanisms to Describe Shock

- There are **five** main measurements used to determine what type of shock a patient is in:
 - 1. Post Capillary Wedge Pressure**
 - Measures the pressure in the **left atrium**
 - Increases with cardiac failure
 - 2. Cardiac Output**
 - Cardiac Output = stroke volume x stroke rate
 - 3. Systemic Vascular Resistance**
 - Measures the constriction of arterioles in the systemic vasculature
 - 4. Saturation of Venous O₂ (SVO₂)**
 - Measurement of amount of oxygen in the venous circulation
 - Assess the ability of tissues to **extract** oxygen from the arterioles
 - 5. Central Venous Pressure**
 - Assesses the amount of blood returning to the heart
 - Measures the amount of blood in the Right Atrium

- Lectures to Reference:
 - Lecture 17: Hemodynamics and Circulation
 - Lecture 28-29: Physiology of Cardiac Cycle Parts 1, 2

Total Peripheral Resistance

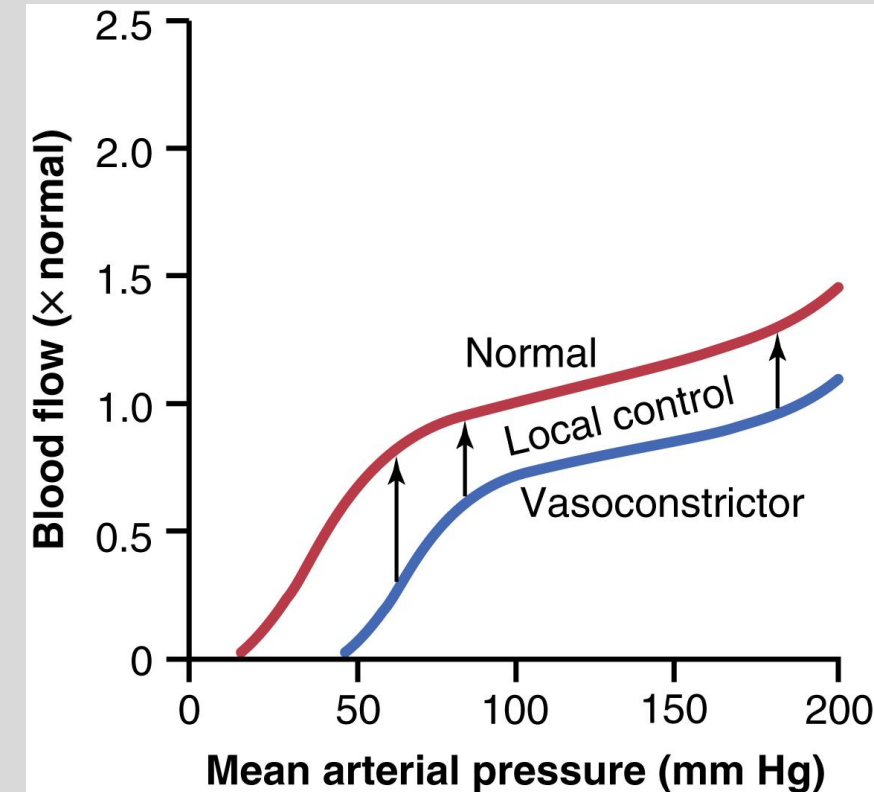
Systemic Vascular Resistance

- Measures the overall level of constriction in the arteriole system
- TPR/SVR is dictated by the resistance of the blood moving through arteries (Poiseuille's Law):

$$R = \frac{8Ln}{\pi r^4}$$

r = radius; L = length of vessel; n = viscosity of blood

- **Radius has the largest effect on TPR**
- **This is an important concept when we consider the effects of vasoconstriction!**



Definition of Shock

- In general, shock occurs whenever there is a generalized, severe reduction in blood supply to the body tissues, and the needs of the tissues are not met
 - Even with cardiovascular compensatory mechanisms, arterial pressure is still low
 - Shock is typically due to one of two things: disturbances in cardiac output or systemic vascular resistance
- Adequate oxygen therapy is reliant on the following physiologic parameters: **cardiac output and systemic vascular resistance**
- In terms of blood pressure, think shock if systolic < 90 mmHg or MAP < 65 mmHg

Main Mechanisms of Shock

Types of Shock	Mechanism	Examples of Causes	Primary Insult
Cardiogenic	Impaired cardiac function	LV Systolic failure Valvular disorders Arrhythmias Right Ventricular Failure	↓ CO
Distributive – Vasodilatory	Impaired vasoconstriction	Sepsis Anaphylaxis Neurogenic Pancreatitis	↓ SVR
Hypovolemic	Volume depletion	Hypovolemia Hemorrhage	↓ Volume (↓ CO due to ↓ PCWP)
Obstructive	Impairment of cardiopulmonary blood flow	Pulmonary Embolism Cardiac Tamponade Tension Pneumothorax	↓ CO

**** The Highlighted Measurement is the primary insult**

- (2022) Approach to the Patient with Shock. Harrison's Principles of Internal Medicine.
- Adapted from Shock and Hemodynamics in the CCU with Dr. Brown, Louisville Medical Lecture Series.

Hypovolemic Shock

- Cause: Loss of blood or plasma volume (~20% blood loss)
 - Hemorrhage
 - Severe vomiting or diarrhea
 - Heat stroke leading to fluid depletion
 - Severe burns
 - Third spacing: shift of fluid from the intravascular space to the interstitial space (pancreatitis, cirrhosis, postoperatively)
- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - Flat jugular neck veins
 - Dry mucous membranes

• (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.

• (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology

• (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Hypovolemic Shock: Pathophysiology

- Hypovolemia leads to **REDUCED CARDIAC FILLING** and **REDUCED STROKE VOLUME**
 - **Preload is decreased**
- **Central Venous Pressure:** decreased
 - Reduced overall volume will decrease amount of blood in RA
- **PCWP: decreased**
 - Reduced overall volume will decrease amount of blood in LA
 - Decreased preload
- **Cardiac Output:** decreased (volume depletion)
 - Reduced stroke volume will reduce overall CO
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, increasing vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

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Cardiogenic Shock

- Cause: Decreased myocardial contractility
 - Acute Myocardial infarction
 - Viral or drug induced cardiomyopathy
 - Decompensated CHF (congested heart failure)
 - Acute or end stage valvular disease
 - Dysrhythmia (due to decreased filling time)
- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - Elevated jugular venous pressure
 - Positive hepatojugular reflex
 - S3 gallop
 - Pulmonary rales

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Cardiogenic Shock: Pathophysiology

- Cardiac failure leads to **INCREASED PRELOAD** and **REDUCED CARDIAC OUTPUT**
- **Central Venous Pressure:** increased
 - Will have fluid accumulation behind the heart (**Right Heart/Venous System**)
- **PCWP:** increased
 - Preload is increased, but stroke volume decreased
 - Will have fluid accumulation behind the heart (**LEFT ATRIUM**)
- **Cardiac Output:** decreased
 - Reduced stroke volume will reduce overall CO (decreased ejection fraction)
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, CNS will promote vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

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Obstructive Shock

- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - **Tension Pneumothorax:** absent breath sounds
 - **Cardiac Tamponade:** distant heart sounds, jugular venous distension (JVD), hypotension
 - **Pulmonary Embolism:** dyspnea and shortness of breath

- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
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Obstructive Shock: Pathophysiology

- Obstruction leads to **DECREASED PRELOAD** and **REDUCED CARDIAC OUTPUT**
- **Central Venous Pressure:** increased
 - Will have fluid accumulation behind the heart
- **PCWP: VARIABLE**
 - Pulmonary Obstruction: **decreased**
 - Causes: pulmonary embolism
 - Blood is obstructed **BEFORE** entering the LA
 - Decrease LV Preload and Increase RV afterload
 - Cardiac Constriction: **increased**
 - Causes: Cardiac tamponade
 - The entire heart is constricted, reducing diastolic filling ability and decreasing stroke volume
 - Decrease RV preload and decrease diastolic filling
- **Cardiac Output:** decreased
 - Reduced stroke volume will reduce overall CO (decreased ejection fraction)
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, CNS will promote vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

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- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
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- (2022) Approach to the Patient with Shock. Harrison's Principles of Internal Medicine.

Distributive Shock

- Causes: loss of vascular tone
 - **Sepsis**
 - **Anaphylaxis**
 - **Neurogenic***
- Patient Presentation:
 - Wide pulse pressure
 - Warm and dry skin
 - Loss of sympathetic tone (neurogenic)
 - Anaphylaxis Signs & Symptoms: hives, urticaria, stridor, airway compromise

- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Distributive Shock: Pathophysiology

- Distributive shock is due to **loss of vascular tone**, leading to **DECREASED RESISTANCE**
- **Central Venous Pressure:** decreased
 - Vascular tone will also decrease in the venous system
- **PCWP:** Decreased
 - Increased vasodilation allows more blood to remain in the circulatory
- **Cardiac Output:** variable
 - **Septic & Anaphylaxis:** increased
 - Sympathetic response to the drop of vascular tone
 - **Neurogenic:** decreased
 - No sympathetic response due to injury to the CNS
- **Systemic Vascular Resistance:** decreased
 - Driving physiology change in distributive shock
- **SVO₂:** increased
 - Lower SVR due to vasodilation will increase the speed of blood traveling through capillaries
 - Increased rate of flow will decrease the amount of time gas exchange can occur

- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Sepsis Criteria: SIRS

- SIRS criteria often used in hospital settings to assess systemic inflammation and/or sepsis
- **SIRS** is defined as meeting two of the four criteria:
 - Temperature over 38 or under 36 C
 - HR > 90
 - Respiratory Rate > 20 breaths per minute or partial pressure of CO₂ less than 32 mmHg
 - Leukocyte count > 12000 or less than 4000 OR over 10% immature bands
- **Sepsis:** Life-threatening organ dysfunction caused by dysregulated host response to infection
- **Severe Sepsis:** sepsis with one or more end-organ failures
- **Septic Shock:** hemodynamic instability despite intravascular volume repletion
 - **Defined as** sepsis and vasopressor therapy needed to elevated **MAP > 65 and lactate > 2 mmol/L**

Box 1. SIRS (Systemic Inflammatory Response Syndrome)

Two or more of:

Temperature >38°C or <36°C

Heart rate >90/min

Respiratory rate >20/min or PaCO₂ <32 mm Hg (4.3 kPa)

White blood cell count >12 000/mm³ or <4000/mm³
or >10% immature bands

From Bone et al.⁹

Early Sepsis: qSOFA & NEWS

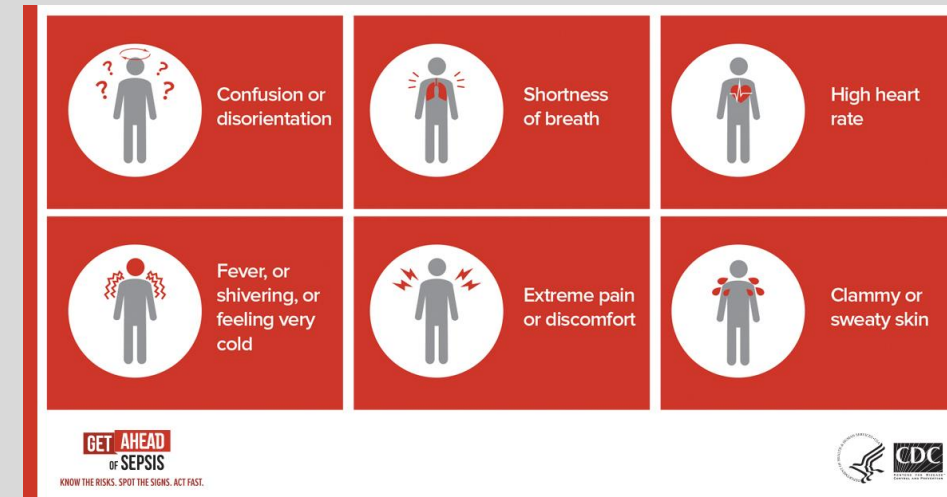
- **Sequential (Sepsis-Related) Organ Failure Assessment (SOFA)**
- **SOFA** is defined as meeting two of the three criteria:
 - Respiratory rate > 22 per minute
 - Altered mentation
 - Systolic BP < 100
- SOFA helps warn clinicians to further investigate organ dysfunction
- **National Early Warning Score**
- Based on a points system of the following physiologic parameters
 - Respiratory rate
 - Oxygen saturation
 - Systolic BP
 - Pulse
 - Level of consciousness/confusion
 - Temperature
- 0-4: low risk
- 5-6: medium risk
- 7+ : high risk

Box 4. qSOFA (Quick SOFA) Criteria

Respiratory rate $\geq 22/\text{min}$

Altered mentation

Systolic blood pressure ≤ 100 mm Hg



Sepsis Treatment:

- Broad spectrum antibiotics while awaiting results of blood cultures:
 - Goal: initiate IV antibiotic therapy within the first hour after diagnosis of sepsis or septic shock
 - Abx include: vancomycin + cefepime or piperacillin
- Hypotension:
 - Begin with IV fluid repletion, particularly crystalloid solutions
 - If fluids are ineffective, can treat with vasopressors
- **Key:** identifying the source of infection and removing it (if foreign body) will increase success rates

Surviving Sepsis
Campaign

BUNDLE

HOURLY-1 BUNDLE: INITIAL RESUSCITATION FOR SEPSIS AND SEPTIC SHOCK:

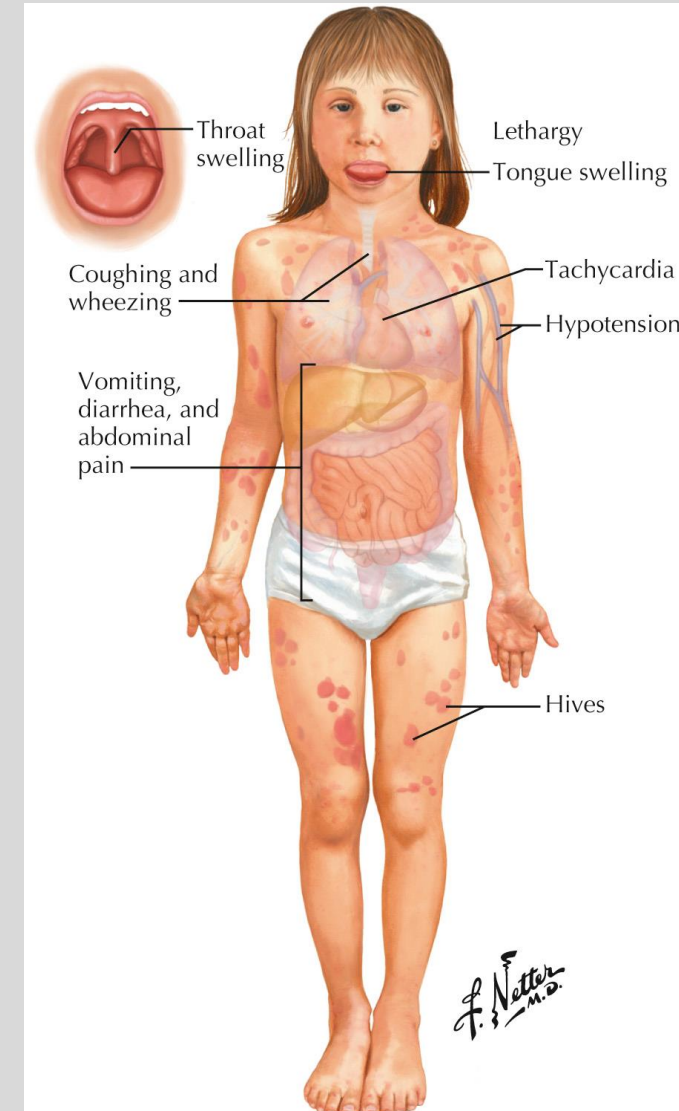
- 1) Measure lactate level.*
- 2) Obtain blood cultures before administering antibiotics.
- 3) Administer broad-spectrum antibiotics.
- 4) Begin rapid administration of 30mL/kg crystalloid for hypotension or lactate ≥ 4 mmol/L.
- 5) Apply vasopressors if hypotensive during or after fluid resuscitation to maintain a mean arterial pressure ≥ 65 mm Hg.

*Remeasure lactate if initial lactate elevated (> 2 mmol/L).

Source: Gregory A. Schmidt, John P. Kress, Ivor S. Douglas:
Hall, Schmidt and Wood's Principles of Critical Care, 5th Edition
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Anaphylactic Shock: Pathophysiology

- Anaphylaxis is an allergic condition where arterial pressure decrease drastically
- Caused by an antigen-antibody reaction
- This reaction causes a cascade of effects, resulting in mast cells in the periphery and basophils in the blood to release **histamine or histamine like-substances**
 - Increase vascular capacity and venous dilation
 - Dilation of the arterioles
 - Increased capillary permeability, with loss of fluid and protein into tissue spaces
- From a physiologic presentation, it is identical to septic shock



- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Anaphylactic Shock: Treatment

- The most important treatment in anaphylaxis is **epinephrine IM**
- Additional treatment should include:
 - Normal Saline to treat hypotension
 - Oxygen
 - Albuterol: help decrease bronchospasm, especially if present after epinephrine

Beneficial effects and adverse effects of epinephrine in the treatment of anaphylaxis

Beneficial effects	
At alpha-1 receptor:	
Increased vasoconstriction (at low doses)	
Increased peripheral vascular resistance	
Increased blood pressure	
Decreased mucosal edema (eg, in larynx)	
At beta-1 receptor:	
Increased heart rate (chronotropy)	
Increased force of cardiac contraction (inotropy)	
At beta-2 receptor:	
Decreased mediator release from mast cells and basophils	
Increased bronchodilation	
Increased vasodilation	
Adverse effects ^{*†Δ}	
Common and transient	Anxiety, palpitations, pallor, tremor, fear, restlessness, dizziness, headache
Uncommon (typically occur after overdose)	Ventricular arrhythmias, angina, myocardial infarction, pulmonary edema, sudden sharp increase in blood pressure, intracranial hemorrhage

* Risk of adverse effects may be increased in the following conditions:

- Use of tricyclic antidepressants, monoamine oxidase inhibitors, or cocaine.
- Some preexisting cardiovascular, central nervous system, or thyroid diseases. Examples include intracranial surgery, acute aneurysm, or untreated hyperthyroidism.

† Serious adverse effects such as those listed in the table potentially occur when epinephrine is given in overdose by any route, most commonly after an intravenous bolus injection, an overly rapid intravenous infusion, or an erroneous intravenous injection of a 1 mg/mL (1:1000) epinephrine solution instead of an appropriately diluted 0.1 mg/mL (1:10,000) or a 0.01 mg/mL epinephrine solution.

Δ Anaphylaxis can present as an acute coronary syndrome, arrhythmias, myocardial infarction, or angina, before or in the absence of epinephrine injection. This potentially occurs in patients with known coronary artery disease, patients in whom subclinical coronary artery disease is unmasked, and patients (including children) with transient coronary artery vasospasm in whom no cardiovascular abnormalities can be detected by electrocardiogram or echocardiography after recovery from anaphylaxis.

Results of Shock

Types of Shock	Examples of Causes	CVP	PCWP	CO	SVR	SVO ₂
Hypovolemic	- Hypovolemia - Hemorrhage	↓	↓	↓	↑	↓
Cardiogenic	- LV Systolic failure - Valvular disorders - Arrhythmias - RV Failure	↑	↑	↓	↑	↓
Obstructive - Decreased LV preload - Increased RV afterload	Pulmonary Embolism	↑	↓	↓	↑	↓
Obstructive Decreased RV preload	Cardiac Tamponade	↑	↑	↓	↑	↓
Distributive Sympathetic drive intact	Sepsis Anaphylaxis - Liver failure - Pancreatitis - Systemic inflammation	↓	↓	↑	↓	↑
Distributive No sympathetic drive	Neurogenic	↓	↓	↓	↓	↑

• (2023) Cardiovascular Function in Pathological Situations. Harris D.M.(Ed.), *Mohrman and Heller's Cardiovascular Physiology, 10th Edition*.
 • (2022) Sepsis and Septic Shock. Harrison's Principles of Internal Medicine.

Management of Shock

Types of Shock	Mechanism	Examples of Causes	Primary Insult	Management
Cardiogenic	Impaired cardiac function	LV Systolic failure Valvular disorders Arrhythmias Right Ventricular Failure	↓ CO	• Inotropes • Further cardiac intervention
Distributive – Vasodilatory	Impaired vasoconstriction	Sepsis Anaphylaxis Neurogenic Pancreatitis	↓ SVR	• Fluid resuscitation • Antibiotics (septic), • IM epinephrine (anaphylaxis)
Hypovolemic	Volume depletion	Hypovolemia Hemorrhage	↓ Volume (↓ CO)	• Fluid resuscitation • Pressors
Obstructive	Impairment of cardiopulmonary blood flow	Pulmonary Embolism Cardiac Tamponade Tension Pneumothorax	↓ CO	• Fluid resuscitation • Relieving the obstruction

Medication: Vasoactive Agents

- While addressing the underlying cause for shock is the best treatment modality, vasoactive agents can help restore blood pressure
- Norepinephrine is a first line agent for septic shock
 - Dobutamine or another inotrope should be used to cardiogenic shock
- Complications:
 - Extravasation: vasoconstriction of nearby connective tissue can cause skin necrosis
 - Hypoperfusion: prolonged vasoconstriction can produce inadequate perfusion of the extremities, mesenteric organs, and kidneys

Medication: Vasoactive Agents

DRUG	Clinical Indication	Receptor Binding	Major Effect	Notes and AE
Norepinephrine	Shock (distributive, cardiogenic, mixed)	$\alpha_1 \gg \beta_1 > \beta_2$	$\uparrow$ SVR $\uparrow$ CO	Usually first-choice pressor AE: Tachyarrhythmias, peripheral (digital) ischemia
Phenylephrine	Shock (distributive, hypovolemic)	α_1	$\uparrow$ SVR	Useful in tachyarrhythmias AE: Reflex bradycardia, severe peripheral and visceral vasoconstriction
Vasopressin	Shock (distributive, cardiogenic)	V_1, V_2	$\uparrow$ SVR	Add to norepinephrine in septic shock AE: Arrhythmias, cardiac ischemia, peripheral and splanchnic vasoconstriction
Epinephrine	Shock (anaphylactic, cardiogenic, distributive), cardiac arrest, bronchospasm	$\alpha_1 > \beta_1 > \beta_2$	$\uparrow$ HR $\uparrow$ SVR $\uparrow$ CO	First line for anaphylaxis and cardiac arrest AE: Ventricular arrhythmias, cardiac ischemia
Dopamine	Bradycardia Shock (cardiogenic, distributive)	$D_1 \gg \beta_1 > \alpha_1 > \beta_2$	$\uparrow$ CO mild $\uparrow$ SVR	Not first-line treatment AE: Ventricular arrhythmias, cardiac ischemia, tissue ischemia/gangrene
Dobutamine	Cardiogenic shock	$\beta_1 \gg \beta_2 > \alpha_1$	$\uparrow$ CO	Inotrope, not vasopressor Can cause hypotension AE: Ventricular arrhythmias, cardiac ischemia

Shock Summary : Cont'd

- Cardiac Shock: cardiac pumping ability is compromised
 - Cardiac output decreases
 - Causes: arrhythmias, valve malfunction, coronary occlusions, myocardial infarction
- Hypovolemic Shock: significant hemorrhage or fluid depletion
 - Decreased circulating blood volume causing decreased cardiac output (decreased stroke volume)
 - Causes: hemorrhage; inadequate fluid intake, severe burns, chronic diarrhea, prolonged vomiting
- Obstructive Shock: reduced cardiac filling due to obstruction or restriction of the heart
 - Decreased stroke volume
 - Causes: cardiac tamponade (prevents adequate diastolic filling); pulmonary embolus (limits blood entering the LA)

Question 1: Answer

An 18 year old female presents to your office severely fatigued. She reports that she has had episodes of diarrhea and vomiting for the past 4 days. Vitals reveal: temperature 102.7°F, BP 75/50mm Hg, and HR 110/min. Hemodynamic measurement reveals a ↑SVR. Which of the following is the most likely diagnosis:

- A. Cardiogenic Shock
- B. Obstructive Shock
- C. Hypovolemic Shock**
- D. Distributive Shock

Rationale:

- Diarrhea and vomiting, the pt is unable to retain or replenish her fluids.
- Elevated temperature: typically can be seen in septic shock and hypovolemic (due to illness)
- Since SVR is elevated, this cannot be septic shock

Question 2: Answer

An 85 year old man presents to the ED with fatigue. He has a past medical history of heart failure, HTN, and gout. History reveals nonadherence with his medications due to difficulty picking up his prescriptions. Vitals reveal: temperature 98.7°F, BP 74/41 mm Hg, HR 127/min, and RR 23/min. His jugular veins are distended. Which of the following hemodynamic measurements do you expect to find on further examination?

- A. ↓CO ↑SVR ↑PCWP
- B. ↓CO ↓SVR ↓PCWP
- C. ↓CO ↑SVR ↓PCWP
- D. ↑CO ↑SVR ↓PCWP
- E. ↑CO ↑SVR ↑PCWP

Rationale:

- This patient is in cardiogenic shock! We expect to see decreased cardiac output, and reflexive vasoconstriction
- PCWP will be increased, as there is a backup of fluid in the LA due to LV failure
- PMHx: patient has known history of heart failure with a nonadherence to medication
- While obstructive shock (PE) may appear similar, PMHx points to heart failure

Question 3: Answer

A 56-year-old man is brought to the emergency department 30 minutes after slicing his arm while chopping wood, leading to significant blood loss. His blood pressure is 92/58 mm Hg. On examination, he appears lethargic and pale. His extremities are cool to the touch.

Choice	Heart rate	Total peripheral resistance	Renin	Cardiac contractility
A	↓	↓	↓	↓
B	↓	↑	↑	↑
C	↑	↓	↑	↑
D	↑	↑	↑	↓
E	↑	↑	↑	↑

Which set of findings would most likely represent the body's physiologic reaction within the first 30 minutes of this patient's injury?

- A. A
- B. B
- C. C
- D. D
- E. E

Rationale:

- This patient is in hypovolemic shock
- We would expect to see a drop in BP, with sympathetic reflexive activation (↑ SVR, ↑ HR)
- This causes increase in **heart rate and contractility; NOT CO**
- Renin will be released in an attempt to raise BP (RAAS System)

Question 4: Answer

A 75-year-old nursing home resident is brought to the emergency department because of confusion. He developed a productive cough with green sputum several days ago. Temperature is 38.9°C (102°F), pulse is 110/min, and blood pressure is 90/60 mm Hg. Results of laboratory studies show a WBC count of 14,000/mm³, a platelet count of 50,000/mm³, and elevated serum lactic acid.

Choice	CVP	PCWP	CO	SVR
A	↓	↓	↓	↑
B	↓	↓	↑	↓
C	↓	↓	↓	↓
D	↑	↑	↑	↓
E	↑	↑	↓	↑

Which set of physiologic parameters is most likely for this patient?

- A. A
- B. B
- C. C
- D. D
- E. E

Rationale:

- This patient is in septic shock
- Primary disturbance is a **DECREASE** in SVR
- This causes increase in heart rate, since we are not losing fluid, we will see an increase in CO
- PCWP and CVP will not increase, the heart is functioning fine

Question 5:

You are an intern on call and are paged to see a 39 year old male. He has a past medical history of cystic fibrosis and had a bilateral lung transplant 2 years ago for which he is taking tacrolimus, azathioprine, and corticosteroids. VS reveal: temp 102°F, BP 70/35 mm Hg, HR 134/min, and respirations 18/min.

What is the primary disturbance causing this patient's shock?

- A. Decreased cardiac output due to decreased cardiac contractility
- B. Decreased cardiac output due to decreased diastolic filling
- C. Loss of fluids due to poor fluid resuscitation
- D. Decrease in systemic vascular resistance caused by reaction to lipopolysaccharide by toll-like receptors
- E. Decrease in systemic vascular resistance due to histamine release caused by autoimmune hypersensitivity
- F. Decrease in systemic vascular resistance due to unopposed parasympathetic tone

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